QQQ Credit Spread Strategy: Backtest Report

Oct 24, 2026 · @Kshitij

Summary

The credit spread strategy does not make money on QQQ after trading costs, in any of the 40-plus versions tested on nine years of minute-level option data (March 2016 to April 2025). The best version lost $552 on a $45,000 account over the whole period; the strategy as specified lost $15,391.

- **Win rates look fine, margins do not** - 69-87% of trades win once the stop is 3x or wider, but the average winner (about $18) is too small against the average loser ($71 to $227, depending on the stop) and the costs on each trade.
- **Costs take a large share of the premium** - commissions, bid/ask spreads and closing costs consumed 25-50% of all credit collected, more in the early years when QQQ traded near $100.
- **The fixes helped but did not get there** - a wider stop, strikes further out, an NDX regime filter, a statistical trade filter, lower-cost execution and a 45-day version each cut the losses; none produced a reliable profit.

Recommendation: do not build this strategy for live trading. The conclusion explains why and what would have to change.

A different approach did hold up. Intraday momentum on QQQ shares earned 13.6% a year from 2016 to 2026 with realistic costs, almost independently of the NDX trader (daily correlation 0.04). It has its own section below and is my recommendation for the second account.

What was tested

The backtest runs the strategy exactly as specified, minute by minute, on real option quotes.

- **Strategy** - sell a QQQ credit spread about 7 days from expiry whenever the 15-minute trend state points up (put spread) or down (call spread) and the 15-minute MACD agrees. Short leg at 10 delta, long leg about 0.7% of the QQQ price further out,

minimum credit 0.04% of the QQQ price. Up to 5 open spreads per side, 30 minutes between entries on the same side.

- **Exits** - stop when the spread's value reaches 2x the credit received (checked every 5 minutes), otherwise close by 15:30 on expiry day. Each trade risks about 1% of the account.
- **Gates** - no new entries when NDX's 20-day realized volatility is above 35% or when VIX is above VIX3M.
- **Data** - every QQQ option quote at one-minute resolution from January 2016 to October 2026 (1,509 expirations), plus a separate set of monthly expirations at five-minute resolution for the 45-day test. The QQQ price for the signals comes from the same quotes.
- **Costs** - entries fill one cent below the mid-market credit; exits pay the full bid/ask spread; standard retail commissions of $0.65 per contract per leg (less for cheap options) plus exchange fees. Quotes with no bid or an implausibly wide ask are never traded against.
- **Period** - March 2016 to April 2025 for all tuning. The final year was held back and was not needed, since no version earned the right to be tested on it.

Two no-signal baselines ran under the same rules for comparison: a weekly put spread and a weekly iron condor, each entered every Friday at 10:00.

Results of the core strategy

Every version of the core strategy lost money over March 2016 to April 2025, and in almost every individual year.

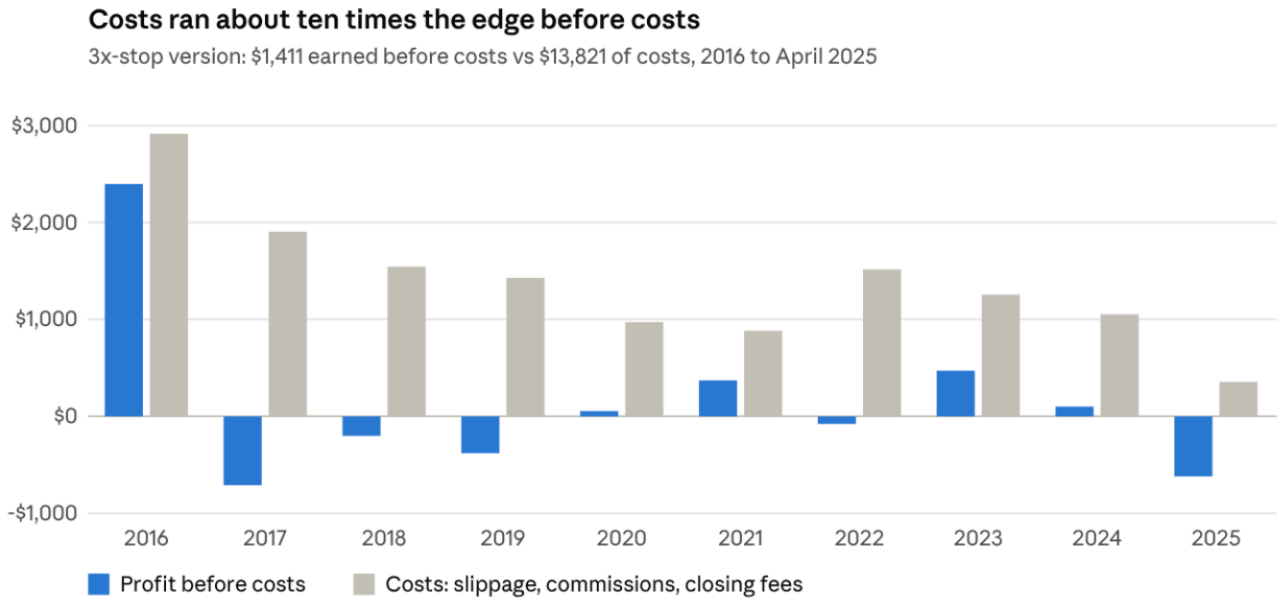
Version	Trades	Win rate	Net P&L	Max drawdown	Profit factor
As specified (2x stop)	1,489	53%	-$15,391	-34.5%	0.49
3x stop	1,317	69%	-$12,410	-28.0%	0.57
5x stop	1,221	80%	-$11,778	-26.7%	0.60
No stop	1,087	87%	-$14,958	-33.8%	0.53
3x stop, let worthless spreads expire	1,370	69%	-$9,726	-22.5%	0.68
Baseline: weekly put spread, no signal	142	73%	-$1,322	-3.3%	0.59
Baseline: weekly iron condor, no signal	415	69%	-$4,327	-9.6%	0.56

Net P&L is in dollars on a $45,000 account; drawdown is the largest fall from a prior peak. Profit factor is gross wins divided by gross losses, so below 1.0 means a loss.

- **The 2x stop fires too often** - with the credit only about 5% of the spread width, a 2x stop sits within normal quote noise and stopped out 47% of trades. Wider stops lose less, but no stop at all is worse than 3x to 5x.
- **Stops filled far above the stop level** - a 2x stop triggered on a gap at the open filled at a median of 2.8x the credit and as much as 7.7x, because the fill is the real price at 09:35, not the stop price.
- **The signal adds nothing per trade** - the trend-timed version and the no-signal condor both lose about $10 per trade. The baselines lose less in total only because they trade far less.
- **Call spreads lose more than put spreads** - in the 3x version, call spreads lost $8,566 against $3,844 for put spreads, with rallies running through the short calls.

Where the money goes

The strategy earned $1,411 before costs over nine years and paid $13,821 in costs, so the problem is a thin edge more than expensive trading.



backtest trade log, 3x-stop version · 1,317 trades, March 2016 to April 2025

Costs fell from 56% of credit collected in 2016 to 25% in 2025 as QQQ rose from about $110 to $510 and each spread collected more premium. The edge before costs did not improve with it: four of the last five years were within $500 of zero before costs.

- **Where the costs went** - commissions $6,111, entry slippage $3,113, exit slippage $2,383, buying back near-worthless spreads at expiry $2,214.
- **Win and loss sizes** - the average winner was $18 and the average loser $71, so the 69% win rate was not enough.

What I tried to fix it

Six changes were tested on top of the core strategy. Each one cut the loss; stacked together they reached breakeven, not profit.

Change	Best result (March 2016 to April 2025)	Effect
Wider stop (3x to 5x the credit)	-$11,778 at 5x	Fewer false stops; the loss falls by a quarter
Strikes further out (5 delta, wider spread)	-$6,739	Fewer losing trades, but each spread collects less, so costs rise to 54% of credit
NDX regime filter: sell only with the regime	-$6,092	Halves the loss and the drawdown by not selling call spreads into bull-market rallies
Statistical trade filter, retrained every two months on the strategy's own past trades	-$4,811 at 5x	Skips about 20% of trades; cuts the loss by more than half
Lower-cost execution (flat per-contract pricing)	-$6,053 (on 2016 to 2020)	Cuts costs from 50% to 31% of credit
All of the above combined (regime filter, 5 delta, 5x stop, low-cost execution)	**-$552**, max drawdown -5.0%, profit factor 1.05	Breakeven: profitable in 5 of 10 years, losses in 2022 and April 2025
45-day expiry instead of 7 (50% profit target, exit at 21 days)	-$7,712	Only positive if every exit fills exactly at mid-market (+$1,617); negative at any realistic fill

- **The regime filter was the single most useful change** - it halved losses at both the 7-day and 45-day expiry.
- **The statistical filter stopped helping once the regime filter and wider strikes were in** - the trades it used to remove were already gone.
- **The combined version is the ceiling** - and it was tuned on the same data it was scored on, so its real expectation is at or below breakeven.

Why the edge is thin

The losses come from the structure of the trade on this market, not from the signal or the costs alone.

- **The protective leg gives most of the premium back** - a spread sold 10 delta out with the long leg 0.7% further out keeps only about 10-15% of the spread's width as credit. What is left is about the size of the bid/ask cost of getting in and out.
- **The account size forces that narrow protection** - keeping each trade's maximum loss within 1-2% of a $45,000 account means the long leg cannot sit far away, so the credit cannot grow.
- **QQQ punishes both sides** - strong rallies run through short call spreads and sharp selloffs (2018, 2020, 2022) through short put spreads. Winners are small and steady; losers are several times larger.
- **Out-of-the-money QQQ options are priced efficiently** - with very liquid markets and tight quotes, the premium on far-out options is close to fair value, leaving little for the seller after costs.
- **The timing signal does not add value here** - trade for trade, the trend-timed entries did no better than entering every week with no signal.

None of this is a flaw in the backtest. The trades filled at real quotes, stops filled at real post-gap prices, and quotes with no real bid were never traded against.

An alternative that holds up: intraday momentum on QQQ

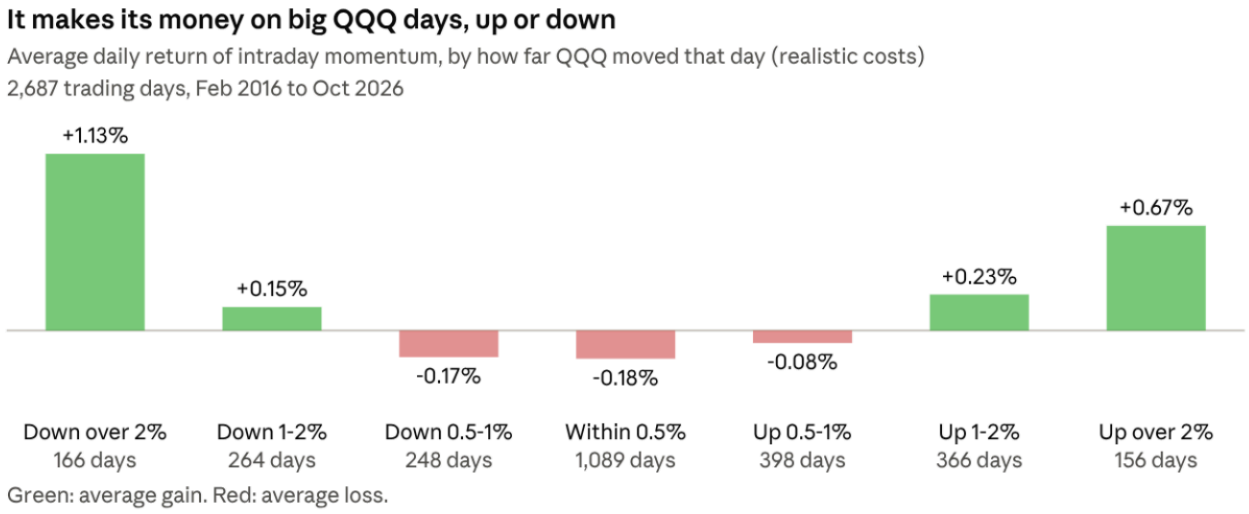
Alongside the credit spreads, I tested intraday momentum on the same QQQ minute data. It trades QQQ shares only, long or short, and holds nothing overnight. Unlike the credit spreads, it makes money after costs, and it behaves almost independently of the NDX trader. The NDX trader figures in this section come from a backtest of its live rules over the same dates.

How it works

- **Normal range** - for each time of day, the average distance QQQ has moved from its open by that time over the last 14 sessions. It forms a band around today's open and yesterday's close.
- **Entries and exits** - at 10:00 and every half hour until 15:30: long if QQQ is above the band and above the day's average price, short if it is below the band and below the day's average price, flat otherwise. Everything is closed at 16:00.
- **Position size** - 2% divided by QQQ's recent daily volatility, capped at 2x the account; the median is 1.9x. It holds a position on 59% of days.
- **Costs** - tested two ways. Base: $0.0035 commission plus $0.001 slippage per share each way, filled at the signal price. Realistic: $0.009 per share each way, which pays half the 1-cent bid/ask spread on every fill, with every order filled one minute after the signal. Figures below use realistic costs unless marked otherwise.
- **Source** - the rule was published in 2024 by Zarattini, Aziz and Barbon. I tested it with the published settings and tuned nothing on this data.

Why it works

Most intraday moves are noise and fade. A move that breaks well outside the normal range tends to keep going into the close. The usual explanation is that option dealers hedging their books, and other flows such as leveraged funds rebalancing near the close, trade in the same direction as the move. The strategy therefore behaves like owning a straddle without paying a premium: it earns on big days in either direction and gives back a little on quiet days.



momentum backtest, realistic costs · QQQ 1-minute prices, Feb 2016 to Oct 2026

- **Big days carry it** - the 322 days when QQQ moved more than 2% either way, 12% of all days, produced more than all of its net profit.
- **Small losses, larger wins** - only 44% of trading days are winners, but the average win (+0.87%) is 1.7 times the average loss (-0.51%).
- **No overnight risk** - QQQ's worst day in the period was -12.5%; the strategy's worst day was -3.4%.

Results

A $45,000 account, February 2016 to October 2026. Cash is assumed to earn nothing.

	Base costs	Realistic costs
Annual return	17.0%	13.6%
Max drawdown	-12.3%	-16.7%
Sharpe ratio	1.38	1.14
Calmar ratio	1.38	0.81
Worst month	-6.1%	-7.1%
Positive years	11 of 11	8 of 11
Last 2 years: annual return, Sharpe	11.3%, 0.89	7.0%, 0.60

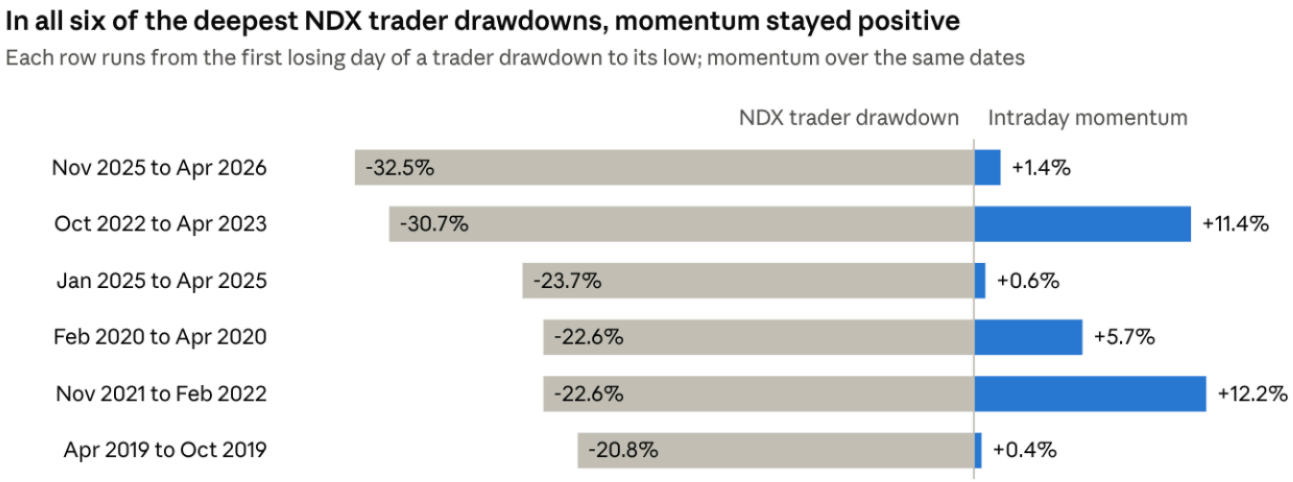
Sharpe is return per unit of volatility; Calmar is the annual return divided by the max drawdown.

Year	Momentum, base costs	Momentum, realistic costs	NDX trader
2016 (from February)	+0.3%	-5.9%	-7.1%
2017	+8.6%	+3.7%	+70.2%
2018	+54.2%	+52.8%	+34.6%
2019	+1.9%	-1.3%	+19.0%
2020	+14.6%	+15.6%	+86.3%
2021	+16.7%	+18.0%	+31.7%
2022	+22.8%	+20.1%	+0.4%

Year	Momentum, base costs	Momentum, realistic costs	NDX trader
2023	+31.1%	+28.7%	+40.6%
2024	+29.5%	+17.6%	+63.0%
2025	+6.9%	-0.1%	+11.7%
2026 (to 8 October)	+5.1%	+6.7%	+16.4%

How it fits with the NDX trader

- **Nearly independent** - the daily correlation with the NDX trader is 0.04, and with QQQ -0.06.
- **It kept earning when the trader lost** - in the 48 months out of 129 when the trader lost money (-5.0% on average), momentum averaged +1.3% and was positive in 62% of them. It earned about the same in the trader's winning months (+1.0%). Both lost in the same month only 18 times.



backtests of the NDX trader's live rules and of momentum with realistic costs · Feb 2016 to Oct 2026

- **Different bets** - the trader bets on direction over days and weeks and holds TQQQ overnight. Momentum bets on the size of each day's move, in either direction, and is flat every night.
- **The trader's bad periods are volatile periods** - selloffs and whipsaws bring large intraday moves that keep going, which is what momentum earns on. In a selloff it shorts the slide, as from November 2021 to February 2022. In a whipsaw each day's swing is a trade for it, as from October 2022 to April 2023, when QQQ rose 18.9% while the trader fell.

- **It cannot offset overnight gaps** - when the trader's loss happens before the open, momentum has nothing to trade, and those days often bounce. That is why it works as a separate source of return rather than a one-for-one hedge.

Running both side by side, as two equal accounts rebalanced to 50/50 each month:

	NDX trader on all capital	Half NDX trader, half momentum
Annual return	31.5%	23.6%
Max drawdown	-32.5%	-16.4%
Sharpe ratio	1.13	1.46
Calmar ratio	0.97	1.44
Worst month	-17.6%	-9.4%

The pair gives up about 8 points of annual return for half the drawdown and half the worst month.

Limits

- **Calm markets earn little** - in sideways months (QQQ within 3% over the month) that were also calm, momentum averaged 0.0% a month, against +2.5% in sideways months with large daily swings. Calm years such as 2016, 2019 and 2025 were weak.
- **It is in a drawdown now** - its deepest drawdown began in May 2025, during a calm stretch, and had not recovered by October 2026.
- **A few big days carry it** - without its 25 best days, the annual return falls to 3.6%.
- **Weaker since publication** - over the last two years the Sharpe ratio was 0.60.
- **Execution has to be automated** - orders go in at each half-hour check, and the account needs margin to short and to reach 2x during the day. $45,000 is above the $25,000 day-trading minimum.

Conclusion and recommendation

The credit spread strategy should not be built for live trading in its current form.

- **It fails the go/no-go tests set at the start** - it does not beat a no-signal baseline per trade, and it does not survive realistic costs. Every version lost money over nine years, including the 45-day variant.

- **The best version is breakeven at most** - a loss of $552 over nine years, tuned on the same data it was scored on. Real-money results would be expected at or below that.
- **Cash would do better** - the account would be better left in cash or T-bills than running this strategy, which carries a 5-35% drawdown for no expected return.

What would have to change for a credit spread strategy to work on QQQ:

1. **A much larger account** - wider protection (or none) is what lets a seller keep the premium, and that needs room for larger per-trade losses than $45,000 allows.
2. **A different source of edge** - the premium itself is not enough on QQQ; the strategy would need a timing signal that genuinely predicts quiet periods, which the trend rules do not.
3. **Near-zero execution costs** - commission-free execution and patient fills at mid-market turn the best version from breakeven to slightly positive, but that gain is too small to rely on.

For the second account, I recommend building intraday momentum instead. With realistic costs it earned 13.6% a year from 2016 to 2026, almost independently of the NDX trader, and running the two side by side halved the worst drawdown. It should be expected to earn little for months at a time in calm markets.

Two ideas to try next as add-ons:

- **Short-volatility ETF in calm markets** - a position in an ETF that is short VIX futures, held only while the VIX futures curve slopes upward. It targets the same insurance premium the credit spreads were after, with one ETF trade instead of option spreads, and it would earn in the quiet sideways months where momentum is flat.
- **Opening-gap fade** - trading against unusually large opening gaps in QQQ during the first half hour. Large gaps often partly reverse early in the session, and the trade would finish before momentum's first check at 10:00.

The backtest engine, the option data (through October 2026) and the trade logs are all retained, so any future variant can be tested within a day.

Appendix: settings and conventions

Item	Setting used
Account	$45,000, fixed sizing base (profits not compounded)

Item	Setting used
Risk per trade	1% of base target, never above 2%; total open risk at most 20%
Signal bars	QQQ 15-minute closes; 1-hour bars anchored to the 09:30 open
Trend lines	Time-series moving averages over 14 and 26 bars
MACD	12, 26, 9 on 15-minute bars; histogram must agree with the trend
Entry window	09:45 to 15:45, each 15-minute close
Expiry	Nearest to 7 calendar days, within 5 to 9
Short leg	Strike nearest 10 delta, out of the money, with a live two-sided quote
Long leg	Nearest strike 0.7% of the QQQ price further out
Minimum credit	0.04% of the QQQ price (about $0.30 at QQQ $750)
Entry fill	Mid-market credit less $0.01 per share
Exit fill	Short leg at the ask, long leg at the bid (full spread paid)
Stop	Spread value at or above 2x the credit, checked every 5 minutes from 09:35; marked at the higher of mid-market and intrinsic value
Expiry-day close	All spreads closed by 15:30
Commissions	$0.65 per contract per leg ($0.50 under $0.10 premium, $0.25 under $0.05), $1 minimum per order, plus $0.03 exchange fees
Quote filter	A quote is used only if it has a bid and the ask is within the larger of $0.10 or one bid above it
Gates	No entries when NDX 20-day realized volatility is above 35% or VIX is above VIX3M
Drawdown overlay	Off for research runs so every year contributes trades

Item	Setting used
Data	1-minute QQQ option quotes with Greeks, January 2016 to October 2026; 5-minute monthly expirations for the 45-day test